

QIDK-Based Garbage Vulnerability Point Monitoring Solution Using Real-Time Image Segmentation

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Abstract—This paper presents a comprehensive Garbage Vulnerability Point (GVP) monitoring solution developed using Qualcomm Intelligent Development Kit (QIDK). The system employs real-time image segmentation to determine critical parameters including garbage area coverage, pixel counts, and dumping patterns. We integrated a camera module with the QIDK Kit to enable real-time image capture and analysis with reduced latency. Multiple deep learning models including YOLOv8, YOLOv11, FCN ResNet50, DDRNet23-Slim, and U-Net were trained and compared. Our results demonstrate that YOLOv11 achieves the best performance with a Mean IoU of 0.75 and F1-score of 0.69. The QIDK NPU achieves approximately 100 ms faster processing compared to CPU, with QIDK v2 showing 200 ms improvement over v1. Field deployment validates the system’s effectiveness for smart city garbage monitoring applications.

I. INTRODUCTION

Urban waste management represents a critical challenge for developing smart cities. Areas prone to garbage dumping pose significant environmental and public health risks to communities. The accumulation of waste creates breeding grounds for disease vectors, generates foul odors, and contributes to environmental degradation. Effective waste management requires comprehensive monitoring of dumping patterns, collection schedules, and spatial distribution of garbage accumulation points.

Traditional manual inspection methods for garbage monitoring are labor-intensive, time-consuming, and lack real-time capabilities. The emergence of edge computing devices and artificial intelligence presents opportunities to automate and enhance garbage monitoring systems. This project addresses these challenges by developing a QIDK-based solution that combines real-time image capture with advanced segmentation algorithms to monitor Garbage Vulnerability Points (GVPs).

The primary objectives of this work are: (1) to develop an automated system for determining garbage area and quantity using image segmentation, (2) to integrate camera modules with QIDK for real-time analysis with reduced latency, (3) to compare multiple state-of-the-art segmentation models, and (4) to demonstrate the advantages of NPU acceleration over CPU-based processing.

II. MOTIVATION

The motivation for this research stems from several critical factors affecting urban environments:

A. Environmental and Health Concerns

Garbage dumping sites are not only detrimental to the environment but also pose serious health hazards to nearby residents. The accumulation of waste leads to soil and water contamination, air pollution from decomposing materials, and creates habitats for disease-carrying organisms.

B. Smart City Development

As cities transition toward smart infrastructure, addressing waste management becomes essential. Effective garbage monitoring systems form a cornerstone of smart city initiatives, enabling data-driven decision-making for municipal services.

C. Pattern Analysis Requirements

Understanding dumping patterns, optimal collection schedules, and spatial distribution of waste requires systematic data collection and analysis. Traditional methods lack the temporal resolution and spatial coverage necessary for comprehensive pattern recognition.

D. Immediate Intervention Needs

Problems such as bad odors, insect breeding spots, and visual pollution require immediate attention. Real-time monitoring systems enable rapid response and intervention, preventing escalation of these issues.

III. METHODOLOGY

A. Dataset Construction

A diverse dataset was compiled from three distinct sources to ensure model generalization:

- 1) **TA-Provided Dataset (1500+ images)**: CCTV footage from two primary garbage dumping locations within the city
- 2) **Custom Campus Dataset (100+ images)**: Images captured from various locations inside and outside campus to prevent overfitting to specific locations
- 3) **Google-Sourced Dataset (100+ images)**: Images representing garbage dumping spots from different regions across the country to enhance geographic diversity

All images were annotated into three distinct classes: Waste, Person, and Animal. This classification enables the system to distinguish between garbage and potential interacting entities.

B. Image Segmentation versus Object Detection

A critical design decision involved selecting between bounding box-based object detection and pixel-level image segmentation. Comparative analysis revealed that bounding box methods overestimate the garbage area by 65.2% due to inclusion of non-garbage pixels within rectangular boundaries. Image segmentation provides pixel-accurate measurements, making it the superior approach for precise area calculation and coverage assessment.

IV. MODELS AND FRAMEWORK

A. Model Selection and Architecture

Five state-of-the-art segmentation models were evaluated using the Qualcomm AI Hub platform:

1) *YOLOv8 Segmentation*: This model combines object detection with instance segmentation capabilities. It offers extremely fast inference speeds and is optimized for real-time processing on edge devices. YOLOv8 employs a unified architecture that performs both detection and segmentation tasks simultaneously.

2) *YOLOv11 Segmentation*: An enhanced version of the YOLO architecture featuring modern optimizations specifically designed for Qualcomm Neural Processing Units (NPUs). YOLOv11 incorporates improved feature extraction and segmentation head designs for higher accuracy.

3) *FCN ResNet50*: Fully Convolutional Network (FCN) with ResNet50 backbone represents one of the most stable and classical semantic segmentation architectures. It provides deep feature extraction through residual connections and has proven reliability across various segmentation tasks.

4) *DDRNet23-Slim*: Deep Dual-Resolution Network (DDRNet23-Slim) is specifically designed for real-time lightweight segmentation. This architecture is particularly suitable for edge devices with constrained computational resources and power budgets.

5) *U-Net*: The U-Net architecture excels at accurate mask generation for irregular shapes, making it particularly suitable for garbage spread patterns. It performs effectively even with limited labeled training samples due to its encoder-decoder structure with skip connections.

B. Model Training Pipeline

The training workflow consists of four primary stages:

- 1) **Data Preprocessing**: Annotated datasets are formatted and prepared for model input
- 2) **Model Training**: Iterative epoch-based training minimizes loss functions
- 3) **Loss Minimization**: Models optimize parameters to reduce prediction errors
- 4) **Post-processing**: Final refinement of segmentation masks and metric calculation

The output includes segmented images, coverage area measurements, and pixel counts for each class.

V. EVALUATION METRICS AND RESULTS

A. Performance Metrics

Two primary metrics were employed for model comparison:

1) *F1-Score*: The F1-score provides a harmonic mean of precision and recall:

$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (1)$$

2) *Mean Intersection over Union (Mean IoU)*: Mean IoU measures the overlap between predicted and ground truth segmentation masks, calculated as:

$$\text{IoU} = \frac{\text{Area of Overlap}}{\text{Area of Union}} \quad (2)$$

B. Model Performance Comparison

Table I presents the quantitative comparison of all evaluated models.

TABLE I
MODEL PERFORMANCE COMPARISON

Model	Mean IoU	F1-Score
YOLOv11	0.75	0.69
YOLOv8	0.72	0.69
FCN ResNet50	0.61	0.66
DDRNet23-Slim	0.50	0.66
U-Net	0.49	0.60

YOLOv11 demonstrated superior performance with the highest Mean IoU of 0.75 and competitive F1-score of 0.69. Based on these results, YOLOv11 was selected for deployment on the QIDK Kit.

VI. IMPLEMENTATION AND DEPLOYMENT

A. Android Application Development

A custom Android application was developed featuring the following capabilities:

- Integration of trained YOLOv11 model
- Image acquisition through camera capture or gallery selection
- Real-time segmentation visualization
- Coverage area and pixel count display
- Processing time measurement and display

The application leverages the QIDK's NPU for accelerated inference, providing users with immediate feedback on garbage detection and quantification.

B. NPU Runtime Analysis

The Neural Processing Unit (NPU) serves as a specialized AI accelerator embedded in the QIDK. Unlike general-purpose CPUs, NPUs are optimized specifically for neural network operations, enabling significant performance improvements for inference tasks.

TABLE II
PROCESSING TIME COMPARISON (MILLISECONDS)

Platform	QIDK CPU	QIDK NPU	Raspi 4
Total Time	2206	2076	17054

C. Processing Time Comparison

Comprehensive benchmarking was conducted comparing QIDK CPU, QIDK NPU, and Raspberry Pi 4 implementations. Table II presents detailed timing breakdowns.

The NPU achieves approximately 100 ms faster processing compared to CPU on the QIDK platform. The Raspberry Pi 4 demonstrates significantly higher latency (17.054 seconds total) due to older CPU architecture and lack of optimized machine learning kernels.

Note that for Raspberry Pi 4, the inference CPU time appears high (around 18 seconds) because multiple threads execute concurrently. The wall time measurement reflects actual processing duration (approximately 13 seconds).

D. QIDK Version Comparison

Evaluation of QIDK v1 versus v2 revealed substantial performance improvements in the newer hardware revision. On average, QIDK v2's NPU segments images approximately 200 ms faster than QIDK v1's NPU, demonstrating continued optimization of the hardware platform.

E. Raspberry Pi 4 Implementation

A parallel implementation was developed for Raspberry Pi 4 featuring:

- YOLOv11 model deployment
- Web-based interface accessible through RealVNC
- Local host model execution
- Per-class pixel count visualization
- Processing time display

This implementation provides a cost-effective alternative platform, though with reduced performance compared to QIDK.

F. Field Deployment

The system was successfully deployed in real-world environments for validation. Field testing confirmed the system's capability to operate effectively under varying lighting conditions, weather scenarios, and garbage composition variations.

VII. DISCUSSION

A. Advantages of Image Segmentation

The experimental validation confirms that pixel-level segmentation provides substantially more accurate area measurements compared to bounding box approaches. The 65.2% overestimation observed with bounding boxes would lead to significant errors in garbage quantification and resource allocation decisions.

B. Model Selection Rationale

While YOLOv11 achieved the best performance metrics, the selection process considered multiple factors including accuracy, inference speed, memory footprint, and compatibility with edge hardware. YOLOv11's optimizations for Qualcomm NPUs made it the optimal choice despite only marginal improvements over YOLOv8 in F1-score.

C. Hardware Acceleration Benefits

The substantial performance gap between QIDK and Raspberry Pi 4 (approximately 8x faster) underscores the importance of specialized AI accelerators for edge deployment. The NPU's dedicated architecture for neural network operations enables real-time processing capabilities essential for practical monitoring applications.

D. Dataset Diversity Impact

The multi-source dataset strategy successfully prevented overfitting to specific locations or environmental conditions. Including geographic diversity through Google-sourced images enhanced model generalization across different urban contexts.

VIII. APPLICATIONS AND FUTURE WORK

A. Smart City Integration

The developed system can integrate with existing smart city infrastructure for:

- Automated garbage collection scheduling
- Real-time alert generation for new dumping sites
- Pattern analysis for preventive intervention
- Resource optimization for municipal services

B. Scalability Considerations

The edge-based architecture enables distributed deployment across multiple monitoring points without requiring centralized processing infrastructure. This scalability makes the solution viable for city-wide implementation.

IX. FUTURE DIRECTIONS

The current system demonstrates significant potential for expansion and enhancement in several key areas:

A. Predictive Analytics Based on Human and Animal Activity

The detection of persons and animals in garbage dumping zones opens opportunities for predictive modeling:

- **Dumping Pattern Prediction:** By analyzing the frequency and timing of human presence near garbage spots, the system can predict potential illegal dumping activities and alert authorities proactively
- **Animal Behavior Analysis:** Monitoring animal interactions with garbage sites can help predict disease outbreak risks and identify areas requiring urgent intervention
- **Temporal Pattern Recognition:** Time-series analysis of human and animal activities can reveal peak dumping hours, enabling optimized surveillance and enforcement schedules

B. Enhanced Classification and Waste Categorization

Future iterations can expand the classification system to:

- Distinguish between different types of waste (plastic, organic, electronic, hazardous)
- Estimate volume in addition to area coverage using depth estimation techniques
- Identify specific recyclable materials for resource recovery optimization

C. Integration with IoT Ecosystem

The system can be enhanced through integration with broader Internet of Things infrastructure:

- Multi-sensor fusion combining visual data with odor sensors, temperature monitors, and gas detectors
- Network of distributed cameras for city-wide coverage with centralized analytics
- Integration with municipal waste management systems for automated work order generation
- Mobile alert systems for nearby residents and sanitation workers

D. Scalability and Deployment Enhancements

To facilitate broader adoption:

- Development of solar-powered autonomous deployment units for remote areas
- 5G connectivity integration for real-time streaming and cloud analytics
- Federated learning approaches to improve models while maintaining data privacy
- Cost-effective hardware alternatives for developing regions

system's potential for evolution into a comprehensive waste management intelligence platform.

X. CONCLUSION

This work presents a comprehensive QIDK-based solution for automated Garbage Vulnerability Point monitoring using real-time image segmentation. Through systematic evaluation of five state-of-the-art segmentation models, YOLOv11 was identified as the optimal choice with Mean IoU of 0.75 and F1-score of 0.69. The implementation demonstrates significant advantages of image segmentation over bounding box approaches, with segmentation reducing area estimation errors by 65.2%.

Hardware acceleration through QIDK's NPU provides approximately 100 ms improvement over CPU processing, with QIDK v2 showing additional 200 ms gains over v1. Comparison with Raspberry Pi 4 reveals an 8x performance advantage for QIDK, validating the benefits of specialized AI accelerators for edge deployment.

Successful field deployment confirms the system's practical viability for smart city applications. The solution addresses critical needs for automated garbage monitoring, enabling data-driven waste management strategies and contributing to environmental sustainability and public health improvement. The outlined future directions, particularly predictive analytics based on human and animal activity patterns, demonstrate the